



RS003 Week 7 — English Worksheet

Topic: Fly the Ship in Four Directions · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week your ship comes to life. It **flies** up, down, left, and right while you hold the arrow keys. You learn that the X axis goes across the screen and the Y axis goes down it — **left makes X smaller, right makes X bigger, up makes Y smaller, down makes Y bigger** — and that the ship is drawn wherever those two numbers say.

Words to know: **X axis, Y axis, key press, increase / decrease, speed**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
ship_x = CENTER_X
ship_y = HEIGHT - 100
SPEED = 5
ship_x -= SPEED
```

Does `ship_x -= 5` move the ship left or right? Why?


```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]: ship_x -= SPEED
if keys[pygame.K_RIGHT]: ship_x += SPEED
if keys[pygame.K_UP]: ship_y -= SPEED
if keys[pygame.K_DOWN]: ship_y += SPEED
```

What happens while you hold UP? What about holding DOWN? Why does UP make `ship_y` smaller?


```
pygame.draw.rect(screen, CYAN, (ship_x - 20, ship_y, 40, 50))
pygame.draw.polygon(screen, WHITE, [(ship_x, ship_y - 20), (ship_x - 20, ship_y),
(ship_x + 20, ship_y)])
```

The body and nose both use `ship_x` and `ship_y` . When you press a key, what moves?

2 · Run It 🏃

🎯 Type the example, run it, and fly the ship

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))

SPACE = (10, 12, 40)
CYAN = (80, 220, 255)
WHITE = (255, 255, 255)
RED = (230, 60, 60)

ship_x = CENTER_X
ship_y = HEIGHT - 100
SPEED = 5

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]: ship_x -= SPEED
    if keys[pygame.K_RIGHT]: ship_x += SPEED
    if keys[pygame.K_UP]: ship_y -= SPEED
    if keys[pygame.K_DOWN]: ship_y += SPEED

    screen.fill(SPACE)
    pygame.draw.rect(screen, CYAN, (ship_x - 20, ship_y, 40, 50))
    pygame.draw.polygon(screen, WHITE, [(ship_x, ship_y - 20), (ship_x - 20,
ship_y), (ship_x + 20, ship_y)])
    pygame.draw.rect(screen, RED, (ship_x - 16, ship_y + 50, 8, 10))
    pygame.draw.rect(screen, RED, (ship_x + 8, ship_y + 50, 8, 10))
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Fly in all four directions. Describe how the ship moves for each arrow key:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — Pressing UP should move the ship up, but it moves **down** instead.

```
if keys[pygame.K_UP]:  
    ship_y += SPEED
```

Hint: up means Y gets smaller.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The ship should only move **while a key is held**, but it drifts to the right on its own without any key press.

```
ship_x += SPEED  
if keys[pygame.K_LEFT]:  
    ship_x -= SPEED
```

Hint: the first line runs every loop no matter what.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The keys do nothing at all. The ship never moves.

```
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
    if keys[pygame.K_LEFT]:
        ship_x -= SPEED
```

Hint: `keys` was never asked for. You need `pygame.key.get_pressed()` outside the event loop.

Write the fixed code (show where `keys` is set and used):

Why was it wrong? Why does your fix work?

4 · Modify It

Change the flying speed

Try `SPEED = 2` , then `SPEED = 10` . Run each one.

Write how the flying feels at slow speed versus fast speed:

Move the whole ship together

Make sure every part of the ship — body, nose, and thrusters — uses `ship_x` and `ship_y` , so they all fly together.

Write down two lines that use `ship_x` and `ship_y` :

Hint: if one part forgets to use `ship_x` or `ship_y` , that part stays behind while the rest flies away.

5 · Make It 📷

Build a flying ship

Build a program where:

1. the ship flies in all four directions with the arrow keys,
2. all parts move together,
3. you can choose the flying speed with a `SPEED` constant.

Send a **video** (so the movement is visible), then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I made variables to hold the ship position called ...

Pressing LEFT changes `ship_x` by ... and UP changes `ship_y` by ...

All the parts move together because they use ...

I set the speed to ... because ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video while you fly the ship with the arrow keys. Talk through it like you are teaching someone. Try to use these words: **X axis, Y axis, key press, increase, speed.**

1. Show the ship flying in all four directions as you press keys.
2. Read your four `if keys[...]` lines out loud and say which way each moves.
3. Point at where the ship position is used to draw the body.
4. Change the SPEED live and show the difference.

Write what you will say in your video. Plan it here before you record.

Submit ✓

Send this worksheet + your flying video to teacher on KakaoTalk.